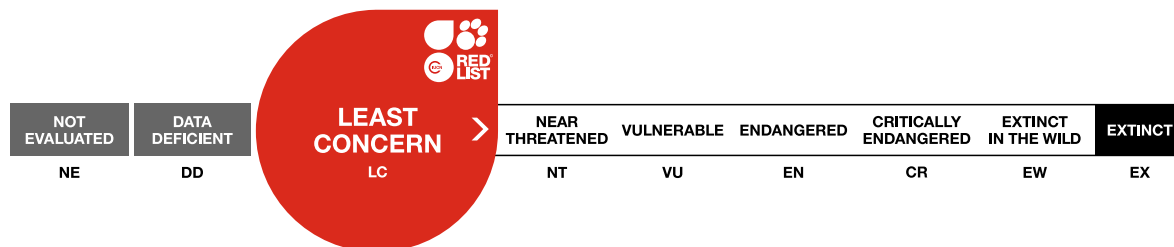


## *Crocodylus porosus*, Saltwater Crocodile

### Errata version

Assessment by: Webb, G.J.W., Manolis, C., Brien, M.L., Balaguera-Reina, S.A. & Isberg, S.



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## Taxonomy

| Kingdom  | Phylum   | Class    | Order      | Family       |
|----------|----------|----------|------------|--------------|
| Animalia | Chordata | Reptilia | Crocodylia | Crocodylidae |

**Scientific Name:** *Crocodylus porosus* Schneider, 1801

### Common Name(s):

- English: Saltwater Crocodile, Estuarine Crocodile, Indo-Pacific crocodile, Salt-water crocodile, Saltie
- French: Crocodile d'estuaire, Crocodile marin
- Spanish; Castilian: Cocodrilo Poroso
- Australian languages: Dungalaba
- Indonesian: Buaya muara
- Malay: Buaya katak
- Palauan: lus
- Papuan (Other): Baya, Kone huala, Pukpuk
- Sinhala; Sinhalese: Geta kimbula
- Tamil: Geta kimbula
- Thai: Jara Kaenumkem

### Taxonomic Source(s):

Uetz, P., Freed, P. and Hošek, J. (eds). 2021. The Reptile Database. Available at: <http://www.reptile-database.org>. (Accessed: 1 June 2021).

## Assessment Information

**Red List Category & Criteria:** Least Concern [ver 3.1](#)

**Year Published:** 2021

**Date Assessed:** November 14, 2019

### Justification:

As a species, the global population of *Crocodylus porosus* is secure because of large populations, extensive habitat and effective management in Australia, Papua New Guinea and, to a lesser degree, Indonesia. There are increasing *C. porosus* populations in the Solomon Islands, Sarawak and Sabah due to effective protection measures, although management may require incentives derived from sustainable use in order to counter negative public attitudes towards them.

### Previously Published Red List Assessments

[1996 – Lower Risk/least concern \(LR/LC\)](#)

1994 – Vulnerable (V)

1990 – Vulnerable (V)

1988 – Endangered (E)

1986 – Endangered (E)

1982 – Endangered (E)

## Geographic Range

### Range Description:

The Saltwater Crocodile (*Crocodylus porosus*) currently occurs in: Australia; Bangladesh; Brunei Darussalam; India; Indonesia; Malaysia; Myanmar; Palau; Papua New Guinea; Philippines; Singapore; Solomon Islands; Sri Lanka; Timor Leste; and, Vanuatu. The species previously existed in Cambodia (possibly extinct); Seychelles (Extinct); Thailand (possibly extinct); and Viet Nam (possibly extinct).

### Country Occurrence:

**Native, Extant (resident):** Australia; Bangladesh; Brunei Darussalam; India; Indonesia; Malaysia; Myanmar; Palau; Papua New Guinea; Philippines; Singapore; Solomon Islands; Sri Lanka; Timor-Leste; Vanuatu

**Native, Possibly Extinct:** Cambodia; Thailand; Viet Nam

**Native, Extinct:** Seychelles

# Distribution Map

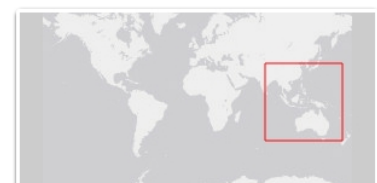
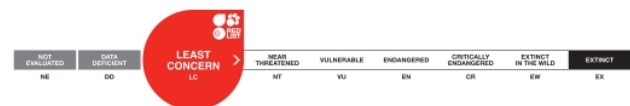


## Legend

■ EXTANT (RESIDENT)

## Compiled by:

Webb, G.J.W., Manolis, C., Brien, M.L., Balaguera-Reina, S.A. & Isberg, S. 2020



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

## Population

The total wild population of *Crocodylus porosus* is estimated to be in excess of 400,000 non-hatchlings. As a species, the global population of *C. porosus* is secure because of large populations, extensive habitat and effective management in Australia, Papua New Guinea and to a lesser degree Indonesia (mainly Papua and West Papua Provinces). There are increasing *C. porosus* populations in the Solomon Islands and Malaysia (Sarawak and Sabah) due to effective protection measures and future management may require incentives derived from sustainable use to counter growing negative public attitudes due to human-crocodile conflict (HCC). Reintroduction and protection efforts in Bhitarkanika National Park, India, have been successful although increasing HCC is now occurring. In contrast, re-establishment of populations outside of protected areas in India may never be possible due to the large human population and lack of suitable habitat. Likewise, re-establishment of wild populations of *C. porosus* in Thailand, Viet Nam and Cambodia, where the species has long been presumed extinct, is unlikely, even in protected areas.

The species has the most commercially valuable skin of any crocodilian (Fuchs 2006) and was intensively hunted for skins historically throughout its range, particularly from the mid-1940s to the 1970s. Restoration of depleted populations have been demonstrated to recover reasonably quickly if: (1) habitats are intact; (2) they are given the opportunity through effective protection; and (3) people are prepared to tolerate expanded populations. Examples are the populations in Australia, Papua New Guinea and Indonesia which have also implemented successful sustainable use models to assist in this recovery. However, elsewhere these preconditions have not been met. Habitat loss continues to be a major problem in many areas occupied by *C. porosus*, and frequently they are killed as pests regardless of whether their skin is used commercially or not. Another significant source of mortality is drowning in fishing nets, especially in river systems where commercial net fishing occurs (eg. northern Australia). Therefore, resourceful incentives will be required to enhance the tolerance of people co-existing with these apex predators with HCC considered the largest risk to their continued abundance.

Information on the current status of *C. porosus* in each range state is summarized as:

### Australia

Surveys by Professor Harry Messel and colleagues established the baseline for crocodile monitoring in northern Australia in the 1970s (e.g. Messel *et al.* 1978-1987). The population was protected (Western Australia 1970; Northern Territory 1971; Queensland 1974) after intensive unregulated hunting commenced in 1945-46. The recovering Australian population was transferred to CITES Appendix II (for ranching) under Resolution Conf. 3.15 in 1985 (Webb *et al.* 1984), and was given an unqualified Appendix-II listing in 1994 (under the old Berne Criteria, as Resolution Conf. 9.24 was only agreed to in 1994, at the same CITES meeting). Australia is considered the most secure long-term stronghold for *C. porosus* throughout its range due to large tracts of relatively uninhabited land and a successful sustainable use program that led to the exportation of nearly 35,000 skins in 2015 (Caldwell 2017).

**Northern Territory:** Spotlight surveys carried out since the early 1970s indicate that in some rivers the population continues to increase, whereas in others the numbers have stabilized (Fukuda *et al.* 2011) despite high levels of legal egg harvest (ranching program) in almost all river systems (Saalfeld *et al.* 2016). The size structure is now biased towards larger animals (>2 m), and total crocodile biomass continues to increase, even in rivers where numbers have stabilized (Fukuda *et al.* 2011). The total

population in the Northern Territory is considered to be near pre-exploitation levels with an average of five non-hatchling crocodiles per km (Webb *et al.* 2000, Fukuda *et al.* 2011, Saalfeld *et al.* 2016). A sustainable use program based on ranching of wild eggs forms the basis of management (Leach *et al.* 2009, Saalfeld *et al.* 2016), combined with a program of problem crocodile removal (Fukuda *et al.* 2014) and a limited wild harvest by landowners. Recognising the public impact of living with a large predator, the Northern Territory government runs a public education program called “Be Crocwise” to reduce HCC (becrocwise.nt.gov.au). Eight farms are currently in operation and crocodiles were estimated to add \$54.3 million to the Northern Territory economy from direct and indirect contributions of employment, capital expenditure and tourism (Ernst and Young 2017). There is no evidence that introduced cane toads (*Rhinella marina*) have impacted on *C. porosus* at all (Fukuda *et al.* 2016). Buffalo are known to have impacted negatively on nesting in some areas in northern Australia, but the significance has not been quantified (C. Manolis, pers. comm. 2019).

**Western Australia:** Major *C. porosus* populations occur in the northwest of the state. Between 1992 and 2012, the population monitoring program was based on an annual aerial survey in the Cambridge Gulf (Ord River, West Arm), with spotlight surveys carried out less regularly in some areas (Mawson 2004). Regular monitoring is currently restricted to an annual spotlight survey of the King River, which has been surveyed fairly consistently since 1989 (1989–1990, 1992–2015, 2017–2019; WMI 2019a). Helicopter surveys indicated a 5.1% per annum (p.a.) population increase in the Ord River (2000–2008) and a 4.1% increase in West Arm (1999–2008) with no signs that the populations were approaching stability (WMI 2008). Recent spotlight surveys of the Ord River in 2019 suggest that the population in the tidal mainstream may be approaching stability, but the population in the upstream non-tidal mainstream continues to increase significantly (WMI 2019b). In 2015, spotlight surveys of the Prince Regent, Hunter and Roe Rivers confirmed healthy breeding populations with a 300% increase in abundance since 1970 (Parke 2015). Cattle grazing is a potential threat to some nesting habitats, and some illegal harvesting of eggs is known to have occurred in 2009–2010.

The increasing *C. porosus* population has prompted authorities to implement a public safety program, similar to the “Be Crocwise” program in the Northern Territory (Parke 2017). Legal harvesting of juveniles, sub-adults, adults and eggs was undertaken in West Arm between 1989 and 1994 to provide stock for crocodile farms. However, only one farm is currently in operation, in Broome, and it is based solely on captive breeding.

**Queensland:** There is a high-density human population on the east coast of Queensland and significant habitat alteration for agriculture. The abundance of *C. porosus* varies between river systems, with the highest densities of crocodiles reported from northwest Cape York Peninsula (Read *et al.* 2004a, b). The most recent surveys (2009–2010) of 48 major waterways (735 km surveyed) along the east coast of Queensland revealed a relative density of  $0.49 \pm 0.72$  non-hatchlings/km (Queensland Parks and Wildlife Service 2010). For those individuals where size could be estimated (N=258), crocodiles ranged from hatchlings to 3.8 m, with 73% less than 2 m (13% were hatchlings). The number of non-hatchlings could be an under-estimate as 33% of crocodiles counted were “eyes only” which does not allow length to be estimated but tend to be larger, more wary crocodiles. On western Cape York Peninsula, surveys based in 10 key river systems in 2008 showed a relative density of 0.85 non-hatchlings/km with successful nesting occurring (Big Gecko, pers. comm. 2009). Identifiable threats to nesting success include feral pests (pigs) and invasive weeds.

Over the last ten years authorities have been under increasing public pressure to deal with increasing numbers of saltwater crocodiles in urban areas, which seems to be in response to more larger crocodiles moving around the coast, an increasing human population, and increased publicity and the ability to easily report sightings (Brien *et al.* 2017). The highest number of attacks, sightings, removals and relocations have occurred along the populated east coast between Townsville and the Daintree River during wet season months (November-February; Brien *et al.* 2017). There were 35 crocodile attacks in Queensland between 1971 and 2015 (total 0.8 per year; fatal 0.3 per year), mostly involving local people or regular visitors (77.1%), specifically adult males (71.4%; mean age 44 years; Brien *et al.* 2017). There has been an increase in the rate of crocodile attacks over time, with an average of 1.3 per year since 1996, most of which were non-fatal (84%; Brien *et al.* 2017).

In response, in 2017, the Queensland Government embarked on a 3-year monitoring program involving systematic spotlight and helicopter surveys in carefully selected, previously surveyed river systems throughout the state. This is the most comprehensive program since 2003, and will be used to determine the size, distribution and abundance and extent of recovery of the *C. porosus* population over time. Currently, the Queensland Crocodile Management Plan ([https://environment.des.qld.gov.au/\\_\\_data/assets/pdf\\_file/0024/87711/wl-mp-croc-manage-plan.pdf](https://environment.des.qld.gov.au/__data/assets/pdf_file/0024/87711/wl-mp-croc-manage-plan.pdf)) consists of six clearly defined zones, from Central to Far North Queensland, with management actions determined by the zone-specific risks of interactions between crocodiles and people. Similar to the Northern Territory problem crocodile program (Fukuda *et al.* 2014), any *C. porosus* that poses a threat to human safety is removed or relocated. So far, 608 crocodiles have been relocated although this number has fluctuated widely between years (range 1-57; Brien *et al.* 2017).

The utilization of crocodiles on Queensland farms is mostly limited to captive breeding, with some farms importing large numbers of Northern Territory and Western Australian sourced eggs and/or juvenile crocodiles. However, an important step towards improving the public perception of crocodile use was attained in late 2018 by a change in legislation (Nature Conservation Act 1994) to allow for a limited wild egg harvest program. Currently (2019) there is only one permit in place for this to occur in Pormpuraaw, northwestern Cape York.

## **Bangladesh**

In Bangladesh, *C. porosus* are restricted to the Sundarbans (Ganges delta). In 1982, four crocodiles and 20 tracks were recorded during daytime (450 km) and nighttime (95 km) surveys. In 2014–2015, daytime surveys in 351 km of river recorded 42 crocodiles (Aziz and Islam 2018). In 2016, 60 crocodiles and 31 tracks were recorded during the day in 1,893 km of river. Protected in 1974, the population was estimated to comprise 150–200 individuals in 1985, 250–300 adult *C. porosus* in 2000, and by 2012 it was considered to not exceed 100–150 adults (Rashid 2013). The reduction in the population is due to a variety of reasons, including an increasing human population, habitat alteration and illegal killing (eg fishermen believe crocodiles compete with them for fish; Manik 2009). In February 1999, 14,300 ha (143 km<sup>2</sup>) or 2.3% of the Bangladesh Sundarbans was declared a wildlife sanctuary where extraction/collection of wildlife resources was banned. The sanctuary includes Kotka-Kochikhadi in East Sundarbans, Notabele in West Sundarbans, and Nikamol in South Sundarbans.

The Karamjal Crocodile Breeding Centre (KCBC) was established in 2000, and in 2016 breeding stock consisted of two females and one male (Anon 2016a). In early 2017, total stock comprised 215 crocodiles (Anon 2017) from this limited founder population. One of the goals of the KCBC is to make

animals available for release into the wild. However, by early 2016, only about 19 *C. porosus* had been released into canals and rivers of the Sundarbans (Anon 2016b, Manik 2009).

### **Brunei**

Little information is available on the status of *C. porosus* in Brunei. Cox (2006) reported low numbers in most rivers and creeks in Brunei Bay (relative density of 0.33 individuals/km). Habitats are largely undisturbed so the biological potential for recovery exists if incentives for the local human population to accept increased numbers of crocodiles can be provided (Ibrahim and Cox 2006). There have been eight confirmed attacks by *C. porosus* on humans since 2006 (CrocBite 2017).

### **Cambodia**

Saltwater Crocodiles are considered to be extinct in Cambodia. It is believed that *C. porosus* were extirpated from Tonle Sap Lake about 30–50 years ago (Platt *et al.* 2006) although some individuals may still exist in coastal areas (Thuok and Tang 1994, Jelden *et al.* 2005, Ouk Vibol, pers. comm. 2017). Very few *C. porosus* are held on the 900+ crocodile farms in Cambodia (Jelden *et al.* 2005) with the local crocodile farming industry being based on the Siamese Crocodile (*C. siamensis*). Hybridization with *C. siamensis* is not a frequent occurrence (Starr *et al.* 2009).

### **China**

The presence of wild *C. porosus* in southern China is currently unknown. Farming operations on the mainland and on Hainan Island may still have some *C. porosus*, imported from range states such as Thailand (Chen 2001, Geng 2001, Li 2001).

### **India**

Saltwater Crocodiles remain in the northeast coastal region of mainland India and the Andaman Islands (Kar 2013, Singh and Kar 2006). A “rear and release” program in Bhitarkanika National Park (BNP), Odisha State, has released more than 2,950 captive-reared juveniles (approximately 1 m long) between 1977 and 2015 and several released females are known to be nesting successfully (Kar 2017). As such, the *C. porosus* population in BNP has increased significantly since the reintroduction program started from 95 sightings in 1976/77 (relative density= 0.87/km) to 1682 sightings in 2017 (relative density= 13.4/km; Kar 2019). The number of nests has increased from five in the mid-1970s, to 65 in 2009, 70 in 2017, to 101 in 2019 (Kar 2009, 2017, 2019). BNP is considered to comprise around 75% of the total population of wild *C. porosus* in India (Kar 2019).

The population increase in BNP has led to increased dispersal and HCC (Padro and Padhi 2019). A team of crocodile husbandry and management staff at the Dangmal/Bhitarkanika Research Facility are now engaged to capture problem *C. porosus* and relocate them back to suitable locations within BNP (Anon 2008a). Previously, groups of captive-bred *C. porosus* were released by the Forest Department in the southern-most part of BNP, primarily as a deterrent strategy for human activities in the mangrove forest (Kar 2007).

Concerns about the potential impact of the Dhamara seaport and Rengali canal developments on BNP and its *C. porosus* population were raised previously (see Webb *et al.* 2010). However, Dhamara seaport has now been in operation since 2010 and does not appear to have had a detrimental effect on the crocodile population or its mangrove habitats. Studies conducted by government and non-government agencies on the impact of the Rengali Dam (upstream Brahmani River) on the flow of fresh water also



highlighted negligible impact on mangrove ecosystems, including those in BNP, and at present there appears to be sufficient freshwater flow from both the Brahmani and other rivers (e.g. Baitrani and Kharasrota Rivers).

In the Indian portion of the Sundarbans in West Bengal, daytime surveys in January 2012 reported a relative density of 0.11 non-hatchlings/km (1,163 km), with a bias towards adults (49.3%) although this is typical of daytime surveys and is not considered to reflect the real population size structure (Choudhury *et al.* 2012). As of March 2012, more than 450 *C. porosus* had been bred at the Forest Department of West Bengal's farm in the Sundarbans and released into the wild (Nagchoudhury 2012).

Although *C. porosus* are known to be widely distributed in areas with suitable breeding habitat (freshwater swamps), there are no recent data on the status of *C. porosus* in the Andaman Islands (Whitaker and Whitaker 1978, Andrews and Whitaker 1994). Human occupation is considered to be displacing crocodiles (Andrews and Whitaker 1994) and HCC is an increasing problem (Whitaker 2008, 2009; Giles 2015) that threatens tourism, which is the biggest industry in the islands. A review of management options for the Andaman Islands was conducted in July 2018 (G. Webb, pers. comm. 2018).

## Indonesia

An extensive survey program was conducted by the Food and Agriculture Organization (FAO) and the Indonesian CITES Management Authority (PHKA - Direktorat Jenderal Perlindungan Hutan dan Konservasi Alam) in the 1990s, but it did not provide an estimate of the total population of *C. porosus* within Indonesia, nor its distribution across the archipelago nation (CITES 1994). The stronghold for *C. porosus* is West Papua and Papua Provinces, particularly the Mamberano River drainage in the north. Past human exploitation and habitat alteration have reduced the population of *C. porosus* throughout much of Indonesia, although localities in Sumatra and the extensive rivers and lowlands of Kalimantan clearly support wild populations. With the exception of Papua and West Papua Provinces, captive breeding and ranching (subject to approval and population monitoring) of *C. porosus* is permitted elsewhere in Indonesia (e.g. Java, Sumatra, Kalimantan, Sulawesi). In July 2007, there were 15 registered farms in Indonesia (Manolis 2007, Manolis and McInnes 2007), but the number is now reduced. The last intensive review of crocodile conservation and management in Indonesia was in the early 1990s (Webb and Jenkins 1991), although some aspects of management were dealt with by a IUCN SSC Crocodile Specialist Group (CSG) mission in 2014 (Brien *et al.* 2014).

Despite a self-imposed hunting moratorium between 1994 and 1996, it seems that by 1997 many crocodile farms in Papua Province had stockpiles of wild *C. porosus* skins collected during this time. Based on these stockpiles, the management plan was revised in 1997 to allow ranching (Directorate General of Forest Protection and Nature Conservation 1997) (H. Kurniati, pers. comm. 2009). A system of ranching (juveniles) and wild harvest (sub-adults), similar to that established in neighbouring Papua New Guinea, is now well established in West Papua and Papua Provinces.

Annual export quotas are established by the Indonesian CITES Scientific Authority (LIPI - Lembaga Ilmu Pengetahuan Indonesia) based on monitoring results, and the CITES Management Authority (PHKA) regulates the quotas. Spotlight surveys, which form the basis for monitoring the wild *C. porosus* population (Kurniati and Rumarbar 1999), were not undertaken between 1998 and 2008, and a zero export quota for the species was put in place. In 2007, it was reported that about 10,000 hatchlings

were taken through ranching (H. Kurniati, pers. comm. 2009). Industry is also interested in an egg harvest, which was undertaken on a trial basis with the New Guinea Crocodile (*C. novaeguineae*) in the early 1990s under an FAO project (Manolis and McInnes 2007), but costs of collection in the remote swamps of West Papua and Papua Province have been prohibitive (H. Kurniati, pers. comm. 2009). Concerns have been raised that the harvest quotas in Papua Province could be increased and that the system of quota allocation between farms may not be equitable (Manolis and McInnes 2007). Although minimum and maximum size limits for wild *C. porosus* skins (25 cm and 51 cm belly width respectively) are established, large skins are still purchased illegally by buyers. It is difficult for hunters to distinguish between the two species of crocodiles around the maximum size limit.

It is widely recognized that the crocodile industry in Indonesia (as in Papua New Guinea) is an important, and sometimes the only, source of cash income for many rural communities in remote areas. In 2015, 8133 skins were exported from Indonesia (Caldwell 2017).

## Malaysia

The Malaysian population of *C. porosus* was transferred to CITES Appendix II in 2016, mainly to allow the State of Sarawak to implement a wild harvest and ranching program to create incentives to address increasing levels of HCC (CITES 2016). A zero export quota for wild specimens of *C. porosus* currently applies to Peninsular Malaysia and Sabah.

**Peninsular Malaysia:** Saltwater Crocodiles were considered relatively rare but are now sighted more regularly throughout Peninsular Malaysia (e.g. Klang River; Kwan 2017). Sebastian (1993) listed 10 localities where *C. porosus* had been reported and suggested that the Setui-Chalok-Bari basin on the east coast near Terengganu may contain the most significant population, but this is yet to be confirmed by surveys. A preliminary survey of *C. porosus* in Rembau Estuary in 2009 indicated a relative density of 2.9 non-hatchlings/km with the presence of hatchlings indicating successful nesting the previous season (Nazli and Hashim 2009). In July 2013, a large *C. porosus*, estimated to be around 6 m long, was found dead on the banks of the Kuala Selangor River at Bukit Belimbing, Kuala Selangor. A sack and plastic bag found in its stomach suggested that pollution may have been implicated in its death (Chen 2015).

**Sabah:** Surveys of the *C. porosus* population have been undertaken irregularly over time in Sabah. Stuebing and Mohammed Sah (1992) surveyed the Klias River and found a small but viable population of around 50 individuals (hatchling-adult), while Cox and Gombek (1985) reported uniformly low densities throughout Sabah. Kaur (2006) reported that the relative density of *C. porosus* in the Segama River had increased from around 0.04 individuals/km in 1981 (Whitaker 1984) to 1.42 individuals/km in 2005. Stuebing (2002) reported a mean density of 1.1 non-hatchlings/km in Sabah rivers—22 times that reported by Whitaker in 1984 (0.05 non-hatchlings/km). A three-year state-wide survey program across 11 of Sabah's largest rivers, selected on the basis of historical data and incidence of HCC, was completed in late 2019 to assess status and population trends. Preliminary results from six of those rivers indicated a mean relative density of 0.66 non-hatchlings/km; Luke Evans pers. comm. 2017). Notwithstanding that the preliminary mean density reported in 2019 is lower than that reported by Steubing *et al.* (2002), there is little doubt that the *C. porosus* population in Sabah has recovered significantly over the last 20+ years. This is considered to be the result of a combination of legal protection (since 1982), a decline in the timber industry which decreased habitat/river disturbance, stabilization of oil palm estates and secondary growth along river banks, opening of closed canopy swamp and riverine forest, the El Niño-Southern Oscillation episode of 1997-98 which reduced flooding of nests and led to high recruitment

that year, and a dramatic decline in the harvest of wild crocodiles for skins during the late 1990s due to both lower prices and the implementation of CITES (Steubing *et al.* 2002).

The increase in HCC in recent years (30 attacks in 2015–2019; CrocBite 2020) is of major concern to authorities (Webb 2008). Problem crocodiles have been captured and removed to farms (Look Fook Soon Trading, unpublished data), although some individuals are shot due to their proximity to human habitation. The current state-wide survey program will inform a new management plan to be drafted in 2020.

**Sarawak:** Saltwater Crocodiles occur in most major rivers in Sarawak and large individuals are sufficiently common to be a serious threat to people. Surveys undertaken in the early 1980s found uniformly low densities of crocodiles throughout Sarawak (0.05 non-hatchlings/km; Cox and Gombek 1985) but recent surveys indicate that numbers have increased markedly since that time (CITES 2016). Surveys conducted in 2012–2014 indicated densities of 0.68 non-hatchlings/km (Sarawak Forestry Department) and 0.83 non-hatchlings/km (Sarawak Forestry Corporation), and the population was estimated to be around 12,000–13,500 individuals (CITES 2016). Sarawak is currently implementing the “Management Plan for Estuarine Crocodile (*Crocodylus porosus*) in Sarawak 2016–2020” which includes removal of problem crocodiles, population surveys and monitoring, wild harvest and ranching, and community-based ecotourism and research.

The size structure of the population, with a high proportion of large individuals, is considered to have contributed to high rates of HCC. Attacks on humans in Sarawak have increased over time (1980–1999: 1.85/yr; 2000–2009: 4.3/yr; 2010–2017: 8.5/yr; CrocBite 2018; Lading 2004) leading authorities to undertaking culling operations in areas where fatal attacks have taken place. In May 1992, a notorious and distinctively marked (white back) *C. porosus*, Bujang Senang, was killed after 30 years in which it is reported to have eaten 13 people (Ritchie and Jong 1993).

## Myanmar

Once widely distributed throughout all coastal areas, *C. porosus* is now largely restricted to the lower Ayeyarwady (= Irrawaddy) River, and coastal Rakhine and Tanintharyi states. The only viable population is in Meinmahla Kyun Wildlife Sanctuary (MKWS) and adjacent forest reserves of the Ayeyarwady Delta, where nesting and recruitment were documented in 1999 (Thorbjarnarson *et al.* 2000; 2003; 2006). The sanctuary was resurveyed in January 2017 and the numbers were comparable to the 1999 survey (S. Platt *et al.* unpublished data). It is speculated that this population may function as a source population for coastal regions in southern Myanmar, including offshore islands in the Myeik Archipelago and one of the only significant populations of *C. porosus* remaining on mainland southeast Asia. Reports of crocodile attacks (CrocBite 2018) indicate that *C. porosus* is present in the Myeik Archipelago, albeit at low densities. MKWS could also be the source of crocodiles that recently turned up near Phuket Island, Thailand.

Platt *et al.* (2015) surveyed Lampi Marine National Park in the Tanintharyi region of Myanmar and found no evidence of an extant crocodile population, although it harboured a population of *C. porosus* as recently as the 1990s. Local extinction is attributed to direct persecution, egg harvesting (for domestic consumption and sale), and capture of crocodiles to sell to buyers from crocodile farms in Thailand. The park appears to be a potential site in which *C. porosus* could be reintroduced. It is also possible the island group could be recolonized by crocodiles dispersing from MKWS or elsewhere on the mainland.

A small population of approximately 20 *C. porosus* is known to occur in Ban Pone Chaung, a mangrove wetland on the mainland near Ywa Thit Village in the southern Tanintharyi region (Platt *et al.* 2014). A few crocodiles reportedly still occur in the Tanintharyi River, although these probably do not constitute a viable population (Platt *et al.* 2012). Other small populations of questionable viability are rumoured to be scattered along the coast from Ywa Thit northwards to Myeik but have not been formally confirmed (Platt *et al.* 2014).

Hatchlings from the MKWS were collected for captive rearing and 68 were released back to the wild when up to 1.2 m long between 1998 and 2001 (Thorbjarnarson *et al.* 2006). A farming-ranching program was implemented by the government in 1978 and the first farm (Thaketa Crocodile Farm) established in 1979 (Aung Moe 1994). Juvenile *C. porosus* collected from the wild (4097 between 1978 and 1999) and captive-bred individuals (3,087 hatchlings between 1983 and 1999) were used to stock the farm. Exports were mainly of live animals to Thailand and Singapore but since 1990 there has been limited production. The farm remains in operation but continues to be poorly managed (Platt *et al.* 2013) due to a range of technical problems (Thorbjarnarson *et al.* 2006). Efforts to register it as a CITES captive breeding facility in early 2008 were unsuccessful. In 2013, the farm maintained 500–600 crocodiles, ranging in size from hatchlings to large adults. Hatching success is poor and may be related to the nutritionally inadequate diet of marine fish and fish skins (obtained from a nearby fish processing facility) provided to adult breeders.

During the 1970s, the Government of Cambodia presented three *C. siamensis* to General Nwe Win, the former ruler of Myanmar. The *C. siamensis* can no longer be accounted for, and it is unknown if these hybridized with the more numerous *C. porosus* in the farm. Platt *et al.* (2013) recommended Thaketa Crocodile Farm concentrate solely on public education where it could make a significant contribution to regional crocodile conservation.

## Palau

Saltwater Crocodiles were extensively hunted in Palau for their skins between the 1960s and 1980s. The first systematic survey of crocodiles undertaken in 1991, by Messel and King (1992b), recorded 42 crocodiles in 112 km of waterways concentrated in two small populations at North Estuary on Belilou and Ngerdok Lake on Babeldaob. Brazaitis *et al.* (2009) estimated the total population to be 500–750 individuals from a survey in 2003. The Bureau of Marine Resources has run a crocodile program since 2004 and conducted surveys between 2005 and 2008. The most recent surveys revealed relative densities of 0.41 to 1.48 individuals/km in coastal mangrove habitat, and 0.41 to 2.81 individuals/km in creeks with the majority of individuals estimated to be 0.6–3.0 m total length (J. Eberdong pers. comm. 2008). The available survey results suggest that the population has not increased significantly since the early 1990s but is probably stable. However, interviews with 46 past and present crocodile hunters in 2002/2003 suggested that there were more crocodiles than there had been historically (Matthews 2003, 2005).

There have been two crocodile attacks reported in Palau, the first in 1965 and the most recent in 2012 (CrocBite 2018). The Palau program focuses on ‘problem’ crocodile complaints and public awareness, with 21 problem crocodiles reported between 2005 and 2008, mostly from the north of the country (J. Eberdong, pers. comm. 2009). A few adult *C. porosus* were maintained in captivity in Koror, which have produced captive-bred hatchlings that have been released back into the wild. The success of

reintroduction remains unknown.

Despite concerns about the genetic integrity of *C. porosus* in Palau due to past importation of different crocodile species (New Guinea Crocodile: *C. novaeguineae*; Philippines Crocodile: *C. mindorensis*, and American Alligator: *Alligator mississippiensis*; Brazaitis *et al.* 2009), an analysis of 39 blood samples from wild *C. porosus* confirmed no hybridization (Russello *et al.* 2007). A CSG review in 2005 highlighted the fact that crocodiles are not currently protected by law, and that the public generally “dislike” them and consider them pests (Anon 2006). At times, crocodiles are killed and eaten as food but are not thought to be detrimental to their numbers.

### Papua New Guinea

*Crocodylus porosus* is widely distributed throughout the lowlands of Papua New Guinea and on the islands of New Britain, New Ireland, Bougainville and Manus. Current management of the wild population involves ranching (eggs, hatchlings, juveniles) and wild harvest (with size limits to protect breeding stock). A monitoring program has included regular nest surveys of representative habitats in the middle Sepik River since 1982 with a review of the program carried out in 1995 and subsequent revision of nesting indices (Manolis 1995). Nesting surveys indicate that the *C. porosus* population in the Sepik River is healthy and increasing at a rate of 2.4% p.a., with greater than 70 nests detected annually from 2003–2014 across 12 primary sites (Solmu *et al.* 2014). This increase has been attributed in part to a conservation awareness campaign and conservation incentives generated from an expanded *C. porosus* egg harvest (Wilken and Langelet 2004, Sine and Kula 2006, Solmu *et al.* 2014). The increasing involvement of local communities in the egg collection program has helped turn a HCC problem into an economic opportunity that supports the conservation of *C. porosus* (Solmu *et al.* 2014). However, increasingly strict skin grading standards in the international market have resulted in reduced purchases of ranched juveniles by farms leading to a reduction in price for wild skins, and thereby threatening the incentives that have been created in some remote areas through the program (E. Langelet, pers. comm. 2018).

Introduced fish (Pacu: *Piaractus brachipomus*, Java Barb (Java Carp): *Barbonymus gonionotus*) continue to degrade nesting habitats (Cox *et al.* 2006, G. Solmu, pers. comm. 2018) although other factors (e.g. climate change) may be implicated. Consideration is currently being given to size and quality of current remaining habitats, land tenure and land use, presence of other fauna and expansion (data collection) into new areas in which the Sepik Wetlands Management Initiative (formed in 1998) is working.

Given the very large area of inaccessible and undeveloped habitat, and the incentives for local traditional landowners to maintain crocodiles, *C. porosus* appears secure in Papua New Guinea. Reports from New Britain suggest that increasing numbers of attacks on people and livestock were due to increasing numbers of *C. porosus* (PNG National Newspaper, 28 August 2009). However, attack data reported by CrocBite (2017) for the country as a whole suggest that the frequency of attacks by *C. porosus* has remained somewhat stable over the last 15 years (5.6 per year; range 1 to 11).

### Philippines

Saltwater Crocodile populations and habitats are reduced throughout the Philippines, and no large populations of *C. porosus* remain (Ortega *et al.* 1994). Today they exist as a few single individuals, small groups, and some iconic populations scattered through remaining wetland habitats. Areas with the highest numbers of *C. porosus* are thought to be on the island of Mindanao (e.g. Ligawasan Marsh),

rivers and estuaries around southern Palawan, Sulu Archipelago in southwestern Philippines, northeastern Mindanao and some part of northeastern Luzon. A recent survey (Manalo *et al.* 2012) found evidence of basking areas that are potential nesting sites in the Agusan River basin, but no large viable population of *C. porosus* is known to exist. The connectivity of the islands of Palawan and Mindanao with the northeast coast of Borneo suggests a shared crocodile population.

The Balabac Group of Islands in southern Palawan had higher numbers of *C. porosus* compared to other areas in the Philippines. Recent population data derived from spotlight surveys at 17 locations reflects that the population in the southern portion of Palawan, at a density of 1.05 individuals/km, is higher than previously thought and the historical declines appear to be reversible. The populations of *C. porosus* in Sulu Archipelago, Zamboanga Peninsula and Siargao Island are still viable (R. Manalo, pers. comm. 2018), and locally protected. Surveys conducted in 2019 recorded the presence of *C. porosus* along the coast of southern Mindanao and in the coastal municipalities of Surigaodel sur Province, with high numbers in Tawi-tawi (1.78 individuals/km) in Sulu Archipelago. Although both *C. porosus* and *C. mindorensis* co-exist in Ligawasan Marsh, interviews with local residents in 2007 suggest that *C. mindorensis* is more commonly sighted than *C. porosus* (Pomares *et al.* 2009). To date, no surveys have been undertaken to quantify the abundance of either species in the area. The northern Sierra Madre National Park in Isabela also appears home to a small population of *C. porosus* (R. Manalo, pers. comm. 2018), which is threatened by killing as pests, by-catch in fishing nets, and agricultural encroachment (Manalo 2004).

In the late 1980s and early 1990s, the majority of the remaining wild adult *C. porosus* population on Palawan (141 individuals) and Mindanao (115 individuals) were caught or acquired from private collectors and relocated to the Crocodile Farming Institute, where they formed the nucleus of a captive breeding population. The progeny of these animals were later relocated to private establishments, some of which have begun commercial farming based on captive breeding (Mercado 2007). Captive-bred stock are thus available for reintroduction efforts in the future (Crocodylus Porosus Philippines Inc., pers. comm. 2018).

Of 26 crocodile attacks in the Philippines in 2000–2015, 20 occurred in southern Palawan (Corvera *et al.* 2017) despite no evidence of an increasing crocodile population. Crocodiles that threaten the safety of people and livestock are usually killed, but in most cases crocodiles are afforded a level of protection by indigenous communities that still practice their cultural beliefs toward crocodiles and their habitats. In addition, continuous information and education campaigns throughout the species' distribution are being conducted to strengthen human-crocodile coexistence (R. Manolo pers. comm. 2019).

In February 2007, a forum convened to address conservation, management and sustainable use of crocodiles in the Philippines (Anon 2007). This led to an exchange of information and experience between organizations in Luzon (Mabuwaya Foundation) and Mindanao (University of Southern Mindanao) (USM Crocodile Research Team 2007; Mabuwaya Foundation Inc. 2007), and closer collaboration with the crocodile farming industry (Crocodylus Porosus Philippines Inc.). In 2015, the Philippine Government reconstituted the Philippine Crocodile National Recovery Team (PCNRT), to be known as the National Committee for the Conservation of Crocodiles (NCCC). The NCCC developed the "Conservation and Management Plan for the Crocodile in the Philippines" that serves as a guiding framework for both species of crocodiles in the country. The key goals and strategies include the establishment of sanctuaries, enhancement of social understanding and acceptance, implementation of

protocols in managing human-crocodile conflict, and effective monitoring and management of wild and captive populations (see Manalo and Mercado 2019). A second forum on crocodiles in the Philippines (theme: Strengthening Partnerships in Conservation Research and Sustainable Management of Crocodiles) was convened in March 2019 (Manalo and Mercado 2019).

### **Seychelles**

Crocodiles were noted when the first Europeans visited the Seychelles in 1609. Re-examination of skeletal material suggests that the species that occurred there at the time of European discovery was *C. porosus*, and not *C. niloticus* as previously assumed (Gerlach and Canning 1993). Europeans had extirpated the population by the early 1800s.

### **Singapore**

Saltwater Crocodiles have been present and breeding in Sungei Buloh Wetland Reserve (SBWR), on the northwest coast of Singapore, since around 2004 (H.C. Beng, pers. comm. 2017). The reserve is around 130 ha in size and was listed as an ASEAN Heritage Park in 2003. Given the high rate of local and international visitation to SBWR, the potential for HCC is of concern to authorities so barriers have been constructed along some walkways that are close to where crocodiles regularly bask to mitigate the HCC risk (H.C. Beng, pers. comm. 2017). CrocBite reports 26 attacks between 1842–1957 with the two most recent attacks in 1974 and 1989. Individual *C. porosus* have been reported in other parts of Singapore. The population was previously estimated to comprise around 15 adults but this number can vary from year to year, and in 2017 only 5–6 adult *C. porosus* were regularly sighted (H.C. Beng, pers. comm. 2017). The source of the *C. porosus* population in Singapore is considered to be Peninsular Malaysia, which is around 1 km from Singapore. One crocodile tagged with a location device by Singaporean authorities regularly travelled every few days between the two countries.

### **Solomon Islands**

The only vessel-based survey of crocodiles in the Solomon Islands was undertaken in 1989 by Messel and King (1990) who identified survey sites based on knowledge from local people and crocodile hunters. Suitable habitat is restricted by the terrain and further reduced by human occupation and agriculture. Most sightings were from three localities - Lauvi Lagoon (Guadacanal), Lake Tatae (Russell Islands) and Ghahirahobo (Santa Isabel). With the cessation of hunting for skins in 1989, the *C. porosus* population has increased, resulting in increased HCC and fatalities in different parts of the country. The situation has been exacerbated by the banning of firearms following civil unrest and the arrival of the Australian-led Regional Assistance Mission (RAMSI) in 2003, such that local people were unable to deal with problem crocodiles themselves.

In 2018 (April–August), WorldFish in partnership with the Solomon Islands Government, conducted a nationwide survey of community knowledge, attitudes, and beliefs toward crocodiles (Van der Ploeg et al. 2019). The purpose was to obtain reliable information on the population status of saltwater crocodiles and the extent of HCC in the country. This would in turn be used to develop a management plan as identified in the Solomon Islands National Biodiversity Strategic Action Plan 2016–2020. A total of 822 people (472 men and 350 women) were interviewed in 234 villages in seven provinces across the Solomon Islands. The knowledge of local people on the number of crocodiles in particular areas proved to be a robust way of estimating abundance. A total of 225 crocodile attacks on people were reported since 1998, 83 of which were fatal (37%) including 31 children. Over the past 10 years, on average, five people per year have been killed by saltwater crocodiles. Since the cessation of commercial hunting in

1993, the saltwater crocodile population has recovered rapidly. It is estimated that there are now between 1,400 and 2,300 non-hatchling Saltwater Crocodiles in the country, with breeding reported on every major island (Van der Ploeg *et al.* 2019).

With more and larger saltwater crocodiles present, the number of attacks on people has and will continue to increase. Throughout the archipelago, communities are taking preventive measures to minimize the risks of crocodile attacks, with people emphasizing the need to supervise children, avoid fishing alone, and be alert at night and during floods. Flashlights are considered an effective tool to prevent a crocodile attack, and in many villages, crocodiles are killed by local hunters with traps and spears as a precautionary measure or in retaliation of an attack. Cultural restrictions on killing and eating crocodiles remain pervasive throughout the country. Crocodiles are widely regarded as taboo (sacred) animals that need to be treated with respect, with attacks often attributed to the wrath of ancestors or to sorcery (Van der Ploeg *et al.* 2019).

### **Sri Lanka**

The status of *C. porosus* in Sri Lanka varies between locations, although how current abundance in different areas compares to historical abundance is unknown. The *C. porosus* population in Sri Lanka was estimated to be around 375 non-hatchlings in 1978, with the majority (70%) confined to the southwest coast of the island (Whitaker and Whitaker 1979). More recently, Amarasinghe *et al.* (2015) conservatively estimated the *C. porosus* population to be around 2000 non-hatchlings, although field surveys are yet to support these numbers. An island-wide survey for crocodylians is underway and may shed light on more accurate and current population sizes in near future. A breeding population is known to exist in Muthurajawela Swamp (Devapriya 2004, Jayawardene 2004), although sporadic breeding is occasionally reported at other locations (e.g. De Silva and de Silva 2008, Gramentz 2008). Surveys in 2007-2008 in the Bentota River revealed a small breeding population, with the size structure strongly biased towards hatchlings (35 cm TL) and yearlings (<70 cm TL: 93%; Gramentz 2008) which is indicative of a depleted population attempting to recover. Devapriya (2004) observed 20 individuals along a 2.8-km stretch of the Dandugam Oya, with 2-9 individuals in 1.7 km of adjacent marsh. *Crocodylus porosus* were also reported in the Madu Ganga (De Silva and De Silva 2008). De Silva (2008) reported an estimated population of 50 individuals in a 5 km stretch of the Nilwala River. Samarasinghe and Chandrasiri, (2013) reported a relative density of 5.1 individuals/km in Bellanwila-Attidiya Sanctuary in 2013. Hatchlings and juveniles have also been observed in Bolgoda Lake, Polwatte River, Matara River, Nilwala River, Walawe River, Hada Oya and Batticaloa lagoon (Samarasinghe pers. comm 2020).

Major threats to the species include the destruction of eggs, killing, fishing and habitat loss (eg urban encroachment, sand mining, conversion for tourism and agriculture) and pollution (De Silva 2008, Samarasinghe and Chandrasiri 2013, Amarasinghe *et al.* 2015). Killing of crocodiles is sometimes “revenge” after attacks on humans but in some cases crocodiles are taken by villagers/fishermen as a source of meat (Madawala *et al.* 2013, Amarasinghe *et al.* 2015). Local communities along the rivers use a variety of crocodile exclusion enclosures for bathing, washing, etc., in response to attacks by *C. porosus* (De Silva 2008). The current practice of relocating “problem” *C. porosus*, sometimes into Mugger (*C. palustris*) habitats, is not considered to be a good long-term strategy that is further exacerbated by authorities lacking expertise with regard to the capture and transport of crocodiles (Amarasinghe *et al.* 2015).

Despite these anthropogenic impacts, the *C. porosus* population is considered to have increased



significantly in Sri Lanka since the late 1970s (De Silva 2013). The total population was estimated to be around 375 non-hatchlings in 1978, with the majority (70%) confined to the southwest coast of the island (Whitaker and Whitaker 1979). More recently, Amarasinghe *et al.* (2015) conservatively estimated the *C. porosus* population to be around 2000 non-hatchlings.

### Thailand

Saltwater Crocodiles are effectively extinct in Thailand. Surveys in the early 1990s revealed sightings of one or two *C. porosus* on Phuket Island (Ratanakorn *et al.* 1994) but the majority of suitable habitat in this area has now been destroyed or occupied by people. Occasional reports of crocodiles such as by local fisherman in the Ranong River, adjacent to the Myanmar border, and a report of a newly-hatched clutch in 2010 (TCMA 2013) suggest a few individuals may still exist. Tamsiripong (2012) reported a small remnant population at Samaesarn Island on the southeast coast and two female *C. porosus* (2.43 and 2.5 m TL) were captured, marked and released in April 2012. Ao Bandon, the largest mangrove inlet on the east coast, may still support a number of *C. porosus* (Ngampongsai and Nabhitabhata 1987) but this remains to be confirmed. TCMA (2013) estimated the wild population of *C. porosus* to comprise around 200 individuals.

As of 31 December 2016, some 156,000 *C. porosus* were held in crocodile farms in the country (Manolis 2017), of which 16 were registered as CITES captive breeding operations at the end of 2017. Farms contain mainly *C. siamensis*, but hybridization between *C. siamensis* and *C. porosus* is well established.

### Timor Leste

The population status of *C. porosus* in Timor Leste is unknown but it is considered to have increased significantly since the country achieved independence in 2002. The largest population exists in Lake Iralalero (Nino Konis Santana National Park, Posto administrative Lautém). In addition, various waterbodies, mainly lagoons and billabongs along the coast, contain crocodiles including Be Malae (Bobonaro), Ai Parapa, Hera, Christo Rei (all Dili), Naktuka and Citrana (all Oecusse), Loes (Liquica), River Irabere and Maurei, Waibani Waiwai, Rubinaha Wai, Matahoi, Luca (all Viqueque), Namalutu, Urunami and Lapalapa, Werukoco, Raumoco (all Lautém), Kasalac, Hasan Foun, Onu Bot, Beco, Raimea no Leolima, Tashilin and Raimea (all Cova Lima), Modomahut (Manufahi), Seical, Vemase (all Baucau), Ribeira Manatuto (Manatuto) and Bican Tidin (Ainaro) (Crocodile Task Force Timor-Leste 2017).

From 2007 to 2019, there have been 59 fatalities and 21 non-fatal attacks on people (CrocBITE 2019) but many are believed to go unreported. In response, the Timorese Government has set up warning signs at known crocodile spots and is regularly visiting affected communities to raise public awareness. Culturally, crocodiles are very important to local people and *C. porosus* is the national animal (Anon 2008b). There is some hunting of crocodiles for meat by coastal communities and some *C. porosus* are held in captivity in the capital, Dili. The Government has constructed a crocodile enclosure in Hera, near Dili, to enable the removal of problem crocodiles. Community-based monitoring was conducted in various communities in Lautém and Viqueque to assess data on crocodile habitat and attacks, integrating the knowledge of local stakeholders. Local authorities raised the concern that crocodiles dispersing from Australia could be responsible for the increase in crocodile attacks. Future research and management should focus on testing the dispersal hypothesis, and on the removal of problem crocodiles from areas where crocodile habitat and human activity frequently overlaps, as well as on developing an ecotourism strategy including “crocodile watching”.

## Viet Nam

Saltwater Crocodiles persisted in southern Viet Nam and the Mekong delta until 20-25 years ago, but extensive habitat degradation and the direct capture and killing of crocodiles greatly reduced the wild population. By the mid-1940s no more than 100 *C. porosus* were thought to survive in the wild (Cuc 1994). No viable wild populations are known to exist today. A re-introduction and recovery program for *C. porosus* in remaining suitable habitat was proposed by a CSG review mission (Jelden *et al.* 2008) but no progress has been made to date. Very few *C. porosus* are on crocodile farms (Jenkins and Sung 1998, Jelden *et al.* 2008, H. Jenkins, pers. comm. 2018). Nonetheless, hybridization of *C. porosus* with either *C. siamensis* or *C. rhombifer* (imported into Viet Nam from Cuba in 1985) is a possibility (Jelden *et al.* 2008).

## Vanuatu

The eastern-most population of *C. porosus* is recorded from eastern Vanua Lava in Vanuatu (New Hebrides and Banks Islands). The area was surveyed in 1992 by Messel and King (1992a), who concluded that crocodiles were on the verge of extinction there. Only two adult crocodiles were seen in the wild and the population was no longer breeding. In 2003, Australia Zoo was asked by the Vanuatu Government to capture and relocate a 3.6 m *C. porosus* from the heavily populated island of Maewo to Vanua Lava where it is believed to have originated. The population comprises perhaps 15–20 individuals that had recovered from a much smaller population after Tropical Cyclone Wendy in late 1972 (F. Hickey, pers. comm. 2018).

## Other

Stray *C. porosus* have been encountered considerable distances from their normal range. For example:

- In 1971, a 3.8 m *C. porosus* was captured in southern Pohnpei, Eastern Caroline Islands, around 1,360 km from the nearest population (Allen 1974, Buden and Haglelgam 2010).
- Around 1959, a 1.0–1.5 m long crocodile, assumed to be *C. porosus*, was killed at Eauripik Island, Yap State in the Federated States of Micronesia. Although not reported in the literature, many residents sighted the animal after it was killed (Buden and Haglelgam 2010). In late 1986, another crocodile, also assumed to be *C. porosus*, was reported from Woleai Atoll, Yap State (Eldredge 1994, Buden and Haglelgam 2010).
- Takashima (1955) reported three crocodiles from Japanese territories; one from Iwo Jima (in 1744), one from Amami-Oshima at the northern end of the Ryukyu Islands (in 1800), and a third from Toyama Bay, on the main Japanese island of Honshu. All three were presumed to be *C. porosus*.
- A vagrant crocodile was reported on Nauru Island, 1,160 km from the nearest *C. porosus* population (Webb 1994).
- In October 2004, an individual *C. porosus* was captured in the Marshall Islands, some 2,000 km from the nearest *C. porosus* population in Papua New Guinea (Manolis 2005).

The first reported *C. porosus* captured in the Maldives was in 1998, and the animal was placed in a local zoo (Ali 2015). In 2007, a crocodile, assumed to be *C. porosus*, was captured, and in April 2009 suspected crocodile tracks were investigated at Gaafu Alifu Atoll (Anon 2009). In January-February 2015 more than 10 saltwater crocodiles were reportedly sighted (Ali 2015), and a 3.1 m individual was captured near L. Isdhoo (Fayaz 2015). The closest population of *C. porosus* is in India, some 430 km away (Bindloss 2015).

**Current Population Trend:** Stable

## Habitat and Ecology (see Appendix for additional information)

*Crocodylus porosus* is considered the largest of the living crocodilians, with reported lengths of up to 6–7 m (Webb and Manolis 1989, 2009; Whitaker and Whitaker 2008). *Crocodylus porosus* preys on people when given the opportunity. It is one of the most widely distributed of all crocodilians, ranging from southern India and Sri Lanka, throughout southeast Asia, east through the Philippines to Micronesia, and down through Indonesia, Papua New Guinea and the Solomon Islands to northern Australia.

Within this distribution, *C. porosus* occupies a wide of wetland habitats from the sea (including thousands of islands) to hundreds of kilometres inland. A great deal of ecological research was carried out in the 1970s and 1980s, particularly in Australia and Papua New Guinea. Contrary to its common name “Saltwater crocodile”, which implies a marine existence, the species commonly inhabits non-tidal freshwater sections of rivers, and inland freshwater lakes, swamps and marshes. Indeed, it thrives in freshwater environments and some of the highest densities of *C. porosus* have been reported from heavily vegetated freshwater swamps without any tidal influence (Webb et al. 1977; 1984). Breeding and recruitment take place principally in rivers with significant freshwater input, or in freshwater swamps (Jelden 1981, Webb et al. 1983, Cox 1985). In the marine environment it mostly inhabits tidal rivers and creeks, where salinity changes with different seasons and distance upstream, but they regularly move around the coast between rivers and occupy offshore islands.

Long-distance voyages at sea occur (Allen 1974, Manolis 2005) with unknown frequency (e.g. Cox 1985; Jelden 1985; Webb et al. 1984, 1987; Messel and Vorlicek 1989; Webb and Manolis 1989, 2009). Historical and recent data suggest that individuals move between range states (e.g. Malaysia-Singapore, Australia-Timor Leste). However, the species demonstrates high site fidelity, with long distance movements usually associated with translocated adults, or sub-adult and sub-dominant crocodiles dispersing in search of a territory. Individual crocodiles are also known to travel long distances between breeding and non-breeding sites (>100 km) as well as to and from seasonal feeding sites (e.g. sea turtle nesting). The exact nature of these movements is not well known.

In the tidal waterways of northern Australia, the movement of *C. porosus* between river systems appears to be related to sex and ontogenetic changes in social status. Radio and satellite tracking of large, male *C. porosus* are now providing insights into the pattern of movement out of rivers onto the coast, between river systems, and within tidal and non-tidal habitats, including homing associated with relocated individuals (Read et al. 2007, Kay 2004, Brien et al. 2008, Fukuda et al. 2019). However, a recent study involving the genetic analyses of *C. porosus* nests along the Northern Territory coast, combined with satellite tracking data from five translocated adult males, revealed that the Coburg Peninsula appears to be a significant geographical barrier to movement and dispersal of *C. porosus* from the east and west of the Northern Territory (Fukuda et al. 2019).

While the species is highly territorial, they are known to congregate at times of the year when resources are abundant. There are multiple reports of large numbers of adult crocodiles congregating around dead cattle floating in rivers of the Northern Territory and Western Australia. At Shady Camp lagoon on the Mary River, Northern Territory, large numbers (>100) of sub-adult and adult crocodiles have been reported congregating at a barrage separating tidal from non-tidal waters when mullet are spawning. A similar occurrence happens on islands and beaches of northern Australia each year during sea turtle

nesting seasons (e.g. green, flatback, etc.).

In the Northern Territory of Australia, the recovery of wild *C. porosus* populations following protection (1971) was carefully documented, providing new information on population dynamics. Increasing numbers of large crocodiles over time have been accompanied by decreasing numbers of small crocodiles (Webb and Manolis 1992, Fukuda *et al.* 2011), which are predated or excluded from rivers and sometimes forced into marginal habitats, including upstream freshwater areas used for recreation by people (Letnic and Connors 2006).

Female *C. porosus* mature at around 2.2–2.5 m (12+ years of age in the wild). They lay their eggs in a mound of vegetation during the annual wet season (October-May; Webb *et al.* 1977, 1983). Mean clutch size in Australia is around 50 eggs at 113 g per egg (Webb *et al.* 1983), whereas in Papua New Guinea it is around 60 eggs at 100 g per egg (Cox *et al.* 2006). Incubation is typically 80-90 days (depending on temperature). As nesting is a wet season activity, loss of nests due to flooding is typically very high (Webb *et al.* 1983; Cox 1985). Some wild females actively defend their nests and loss of eggs to non-human predators is exceptionally low. Lizards (varanids) take eggs, but usually when the eggs are dead and rotten, and can be located by smell. The recent invasion of Cane Toads (*Rhinella marina*) into the Northern Territory has reduced the varanid population which reduces this egg predation risk. The loss of very small numbers of eggs to rodents has been recorded (WMI, unpublished data), and wild pigs have been reported to raid *C. porosus* nests, but the significance of this has not been demonstrated.

**Systems:** Terrestrial, Freshwater (=Inland waters), Marine

## Use and Trade

Saltwater Crocodile populations are legally protected in most range states, but protection alone may be ineffective. Management programs based on sustainable use (ranching, wild harvest, captive breeding) have been successfully implemented in Papua New Guinea, Australia and Indonesia, the three countries that contain the majority of the global population of the species. The possibility of establishing sustainable use programs in Malaysia and the Solomon Islands are being explored.

Farming of *C. porosus*, based on captive breeding, is undertaken in Bangladesh, China, Thailand, Singapore, Malaysia, Myanmar, Philippines, Indonesia, Papua New Guinea and Australia. Stock produced through captive breeding is supplemented significantly through ranching programs (eggs, hatchlings and/or juveniles) in Indonesia, Papua New Guinea and Australia. Ranching is not possible in much of the former range of *C. porosus* because wild populations are severely depleted or extinct. In 2015, there was approximately 71,000 *C. porosus* skins traded internationally (Caldwell 2017).

## Threats (see Appendix for additional information)

**Human-crocodile conflict (HCC)** – The largest risk posed to successful *C. porosus* conservation is the cohabitation of humans with an apex predator. Public awareness and education programs can help alleviate parts of this threat such as “Be CrocWise” (Northern Territory, Queensland and Western Australia in Australia). However, when *C. porosus* and humans are forced to cohabit, and even compete for resources (e.g. fishing), the risk of conflict will increase. Innovative solutions to further mitigate these risks are urgently required for continued conservation success.

**Land use** – *Crocodylus porosus* attack cattle and other livestock as they drink at the water's edge. Pastoralists involved in the Northern Territory of Australia's egg and wild harvests are compensated to some degree for stock losses, and are also provided with incentives to conserve habitats. Elsewhere, retaliation killings occur as crocodiles threaten the livelihoods of local people. Fishermen inadvertently trap and kill crocodiles with fishing equipment (hooks and nets).

**Food** – Unregulated fishing may be depleting fish stocks in some range states, which could have detrimental effects on *C. porosus* populations both from resource competition and increasing interactions leading to HCC.

**Invasive weeds** – Introduced plants for the pastoral industry including Para Grass (*Urochloa mutica*) in northern Australia have a high heat during decomposition which increases the temperature experienced by the developing embryos within the nest. No research into the effects on embryo survival or resultant sex ratios has been conducted to know if detrimental impacts could result. Water Hyacinth (*Eichhornia crassipes*) is also reported as a problem in Sri Lanka (Samarasinghe and Chandrasiri 2013).

**Invasive pests** – No detrimental impacts on Australian *C. porosus* from the introduced Cane Toad (*Rhinella marina*) have been reported so far (Fukuda *et al.* 2016). Feral pigs and buffalo are known to have impacted negatively on nesting in some areas in northern Australia, but the significance has not been quantified (C. Manolis, pers. comm. 2019). In Papua New Guinea, introduced fish species (*Piaractus brachipomus* and *Barbonymus gonionotus*) to degrade nesting habitats (Cox *et al.* 2006, G. Solmu, pers. comm. 2018).

**Habitat destruction and pollution** – Urban encroachment, sand mining, tourism and agriculture threaten habitat modification for *C. porosus* as well as pollutants (Samarasinghe and Chandrasiri 2013, Amarasinghe *et al.* 2015) in some range states.

**Traditional harvesting** – Some traditional harvesting occurs by indigenous peoples throughout the range states but poses little threat to the overall *C. porosus* populations.

## Conservation Actions (see Appendix for additional information)

The IUCN SSC Crocodile Specialist Group (CSG) Action Plan (Webb *et al.* 2019), has proposed the following conservation actions for *C. porosus* as high priorities.

1. **Status surveys:** Current quantitative and qualitative assessments of the current status of wild *C. porosus* populations in the majority of range states are lacking. In particular, priorities should be (in alphabetical order) Bangladesh, Brunei, Cambodia, Palau, Timor Leste, Indonesia (outside Papua Province) and Viet Nam.

2. **Management Programs:** Once populations have been assessed, management plans are required, dependent upon survey findings, and designed specifically for the (sub) region. For some populations, the conservation problem appears to have been largely solved (depleted populations are now increasing) but as a consequence HCC may also be increasing. If realistic management programs, tailored to local circumstances, are not developed and implemented, there will be no public or political tolerance for the expanded population. While removal of problem crocodiles combined with public

education have been successful historically, research into alternative strategies and innovations is now required to reduce the threat to humans.

**3. Sustainable use programs:** These programs have, in the past, proved to be a conservation success in recovering the Australia and Papua New Guinea *C. porosus* populations. However, probability of success depends on many factors including international skin market price fluctuations and the capacity of government regulation. An example where ranching may be of benefit is in Sarawak and Sabah (Malaysia) where farming operations are already established, so ranching would provide an avenue for indigenous investment. However, before this can be considered, surveys need to inform the degree to which any harvest could be sustainable.

**4. Review of crocodile management in Indonesia:** Population monitoring of *C. porosus* in Papua and West Papua Provinces has not been undertaken since 1998 (Kurniati and Rambarar 1999). Although wild harvest quotas have been reduced because of no survey data, the allocation of the quota among licensed farms is of concern to some operators (see Manolis and McInnes 2007). The regulatory system may benefit from external review by the CSG or other competent groups. Protection is nominally afforded to *C. porosus* in areas other than Papua Province (CITES 1994) but it seems ranching is occurring in Sumatra, Kalimantan and perhaps other locations. If so, it needs to be sanctioned by CITES, through a proposal to the CITES Secretariat.

**4. Increased regional cooperation in Southeast Asia:** The close association between production and trade in crocodilian products (including *C. porosus* but mainly *C. siamensis*) between Cambodia, Viet Nam, Thailand and China indicates that management, conservation and regulatory activities need to be coordinated in this sub-region (see Jelden *et al.* 2005; 2008). Technical exchange (e.g. training), surveys, development of management programs and mutually supporting regulatory structures (e.g. enforcement, Customs) are recommended.

**5. Crocodile conservation program in India:** An evaluation of restocking and the identification of additional release sites are in need of review. There is an excess of animals now in captivity and insufficient release sites for them. The issue of HCC in Bhitarkanika National Park and adjacent areas needs to be assessed in view of the increasing attacks on local people.

## Credits

|                                        |                                                                                                                                                                                                                                                                                                        |
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## External Resources

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## Appendix

### Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Habitat                                                                                                     | Season              | Suitability | Major Importance? |
|-------------------------------------------------------------------------------------------------------------|---------------------|-------------|-------------------|
| 5. Wetlands (inland) -> 5.1. Wetlands (inland) - Permanent Rivers/Streams/Creeks (includes waterfalls)      | Resident            | Suitable    | Yes               |
| 5. Wetlands (inland) -> 5.2. Wetlands (inland) - Seasonal/Intermittent/Irregular Rivers/Streams/Creeks      | Non-breeding season | Suitable    | No                |
| 5. Wetlands (inland) -> 5.3. Wetlands (inland) - Shrub Dominated Wetlands                                   | Resident            | Suitable    | Yes               |
| 5. Wetlands (inland) -> 5.4. Wetlands (inland) - Bogs, Marshes, Swamps, Fens, Peatlands                     | Resident            | Suitable    | Yes               |
| 5. Wetlands (inland) -> 5.5. Wetlands (inland) - Permanent Freshwater Lakes (over 8ha)                      | Resident            | Suitable    | Yes               |
| 5. Wetlands (inland) -> 5.6. Wetlands (inland) - Seasonal/Intermittent Freshwater Lakes (over 8ha)          | Non-breeding season | Suitable    | No                |
| 5. Wetlands (inland) -> 5.7. Wetlands (inland) - Permanent Freshwater Marshes/Pools (under 8ha)             | Resident            | Suitable    | Yes               |
| 5. Wetlands (inland) -> 5.8. Wetlands (inland) - Seasonal/Intermittent Freshwater Marshes/Pools (under 8ha) | Non-breeding season | Suitable    | No                |
| 9. Marine Neritic -> 9.10. Marine Neritic - Estuaries                                                       | Resident            | Suitable    | Yes               |
| 12. Marine Intertidal -> 12.1. Marine Intertidal - Rocky Shoreline                                          | Resident            | Suitable    | Yes               |

### Use and Trade

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| End Use                                | Local | National | International |
|----------------------------------------|-------|----------|---------------|
| 1. Food - human                        | Yes   | Yes      | Yes           |
| 3. Medicine - human & veterinary       | Yes   | No       | No            |
| 10. Wearing apparel, accessories       | No    | Yes      | Yes           |
| 13. Pets/display animals, horticulture | No    | Yes      | Yes           |
| 14. Research                           | No    | Yes      | Yes           |
| 16. Establishing ex-situ production *  | No    | Yes      | Yes           |

### Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Threat                                                                                                                                         | Timing    | Scope                                                                                                                                                        | Severity            |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1. Residential & commercial development -> 1.1. Housing & urban areas                                                                          | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance |                     |
| 1. Residential & commercial development -> 1.2. Commercial & industrial areas                                                                  | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance |                     |
| 2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.3. Agro-industry farming                                       | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance |                     |
| 2. Agriculture & aquaculture -> 2.4. Marine & freshwater aquaculture -> 2.4.2. Industrial aquaculture                                          | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance |                     |
| 3. Energy production & mining -> 3.2. Mining & quarrying                                                                                       | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance |                     |
| 4. Transportation & service corridors -> 4.1. Roads & railroads                                                                                | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance |                     |
| 5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.3. Unintentional effects: (subsistence/small scale) [harvest] | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                                                                                |                     |
| 5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.4. Unintentional effects: (large scale) [harvest]             | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                                                                                |                     |
| 5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.5. Persecution/control                                        | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                                                                                |                     |
| 8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.1. Unspecified species   | Ongoing   | Minority (<50%)                                                                                                                                              | Negligible declines |
|                                                                                                                                                | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance |                     |

|                                                                                                                                                                               |                                                                                                                                                                                                                                                                |         |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------|
| 8. Invasive and other problematic species, genes & diseases<br>-> 8.1. Invasive non-native/alien species/diseases -> 8.1.2.<br>Named species ( <i>Brachiaria mutica</i> )     | Ongoing                                                                                                                                                                                                                                                        | Unknown | Unknown             |
| Stresses:                                                                                                                                                                     | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.3. Indirect ecosystem effects<br>2. Species Stresses -> 2.3. Indirect species effects -><br>2.3.6. Skewed sex ratios -> 2.3.7. Reduced reproductive success                   |         |                     |
| 8. Invasive and other problematic species, genes & diseases<br>-> 8.1. Invasive non-native/alien species/diseases -> 8.1.2.<br>Named species ( <i>Sus scrofa</i> )            | Ongoing                                                                                                                                                                                                                                                        | Unknown | Unknown             |
| Stresses:                                                                                                                                                                     | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.3. Indirect species effects -><br>2.3.7. Reduced reproductive success                                                    |         |                     |
| 8. Invasive and other problematic species, genes & diseases<br>-> 8.1. Invasive non-native/alien species/diseases -> 8.1.2.<br>Named species ( <i>Bubalus bubalis</i> )       | Ongoing                                                                                                                                                                                                                                                        | Unknown | Unknown             |
| Stresses:                                                                                                                                                                     | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.2. Species disturbance<br>2. Species Stresses -> 2.3. Indirect species effects -><br>2.3.7. Reduced reproductive success |         |                     |
| 8. Invasive and other problematic species, genes & diseases<br>-> 8.1. Invasive non-native/alien species/diseases -> 8.1.2.<br>Named species ( <i>Piaractus brachipomus</i> ) | Ongoing                                                                                                                                                                                                                                                        | Unknown | Negligible declines |
| Stresses:                                                                                                                                                                     | 1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.3. Indirect species effects -><br>2.3.7. Reduced reproductive success                                                                                                          |         |                     |
| 8. Invasive and other problematic species, genes & diseases<br>-> 8.1. Invasive non-native/alien species/diseases -> 8.1.2.<br>Named species ( <i>Rhinella marina</i> )       | Ongoing                                                                                                                                                                                                                                                        | Unknown | No decline          |
| Stresses:                                                                                                                                                                     | 1. Ecosystem stresses -> 1.3. Indirect ecosystem effects                                                                                                                                                                                                       |         |                     |
| 8. Invasive and other problematic species, genes & diseases<br>-> 8.1. Invasive non-native/alien species/diseases -> 8.1.2.<br>Named species ( <i>Barbonymus gonionotus</i> ) | Ongoing                                                                                                                                                                                                                                                        | Unknown | Negligible declines |
| Stresses:                                                                                                                                                                     | 1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.3. Indirect species effects -><br>2.3.7. Reduced reproductive success                                                                                                          |         |                     |
| 9. Pollution -> 9.4. Garbage & solid waste                                                                                                                                    | Ongoing                                                                                                                                                                                                                                                        | Unknown | Unknown             |
| Stresses:                                                                                                                                                                     | 1. Ecosystem stresses -> 1.2. Ecosystem degradation<br>2. Species Stresses -> 2.1. Species mortality                                                                                                                                                           |         |                     |

## Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Conservation Action in Place     |
|----------------------------------|
| In-place research and monitoring |
| Action Recovery Plan: Yes        |

|                                                               |
|---------------------------------------------------------------|
| <b>Conservation Action in Place</b>                           |
| Systematic monitoring scheme: Yes                             |
| In-place land/water protection                                |
| Conservation sites identified: Yes, over part of range        |
| Percentage of population protected by PAs: 25-50              |
| Area based regional management plan: Yes                      |
| Occurs in at least one protected area: Yes                    |
| Invasive species control or prevention: Unknown               |
| In-place species management                                   |
| Harvest management plan: Yes                                  |
| Successfully reintroduced or introduced benignly: Yes         |
| Subject to ex-situ conservation: No                           |
| In-place education                                            |
| Subject to recent education and awareness programmes: Yes     |
| Included in international legislation: Yes                    |
| Subject to any international management / trade controls: Yes |

## Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Conservation Action Needed                                  | Notes |
|-------------------------------------------------------------|-------|
| 2. Land/water management -> 2.1. Site/area management       | -     |
| 4. Education & awareness -> 4.3. Awareness & communications | -     |

## Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Research Needed                                             | Notes |
|-------------------------------------------------------------|-------|
| 1. Research -> 1.2. Population size, distribution & trends  | -     |
| 2. Conservation Planning -> 2.2. Area-based Management Plan | -     |
| 3. Monitoring -> 3.1. Population trends                     | -     |

## Additional Data Fields

|                                                                   |
|-------------------------------------------------------------------|
| <b>Distribution</b>                                               |
| Estimated area of occupancy (AOO) (km <sup>2</sup> ): 8135        |
| Continuing decline in area of occupancy (AOO): No                 |
| Extreme fluctuations in area of occupancy (AOO): No               |
| Estimated extent of occurrence (EOO) (km <sup>2</sup> ): 26749859 |
| Continuing decline in extent of occurrence (EOO): No              |
| Extreme fluctuations in extent of occurrence (EOO): No            |
| Number of Locations: 15                                           |
| Continuing decline in number of locations: No                     |
| Extreme fluctuations in the number of locations: No               |
| Lower elevation limit (m): 0                                      |
| Upper elevation limit (m): 500                                    |
| <b>Population</b>                                                 |
| Number of mature individuals: 500,000                             |
| Continuing decline of mature individuals: No                      |
| Extreme fluctuations: No                                          |
| Population severely fragmented: No                                |
| No. of subpopulations: 50-500                                     |
| Continuing decline in subpopulations: No                          |
| Extreme fluctuations in subpopulations: No                        |
| All individuals in one subpopulation: No                          |
| No. of individuals in largest subpopulation: 100000-200000        |
| <b>Habitats and Ecology</b>                                       |
| Continuing decline in area, extent and/or quality of habitat: Yes |
| Generation Length (years): 25                                     |
| Movement patterns: Not a Migrant                                  |
| Congregatory: Congregatory (and dispersive)                       |

## Errata

**Errata reason:** This errata version of the assessment was created to correct the information that this species was previously thought to be extinct in Singapore: the species has never been considered extinct in Singapore.

## The IUCN Red List Partnership



The IUCN Red List of Threatened Species™ is produced and managed by the [IUCN Global Species Programme](#), the [IUCN Species Survival Commission \(SSC\)](#) and [The IUCN Red List Partnership](#).

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